Code Book: Emotion Annotation on Product Reviews

In this task, you will annotate sentences extracted from product reviews. Your goal is to identify various key facets of emotions, namely the polarity, arousal, emotion tag, emotion cues, emotion source, emotion cause and emotion experiencer. Each of the tasks are detailed below.

Section 1

This section is intended to serve as a guideline to annotate the fundamental emotion facets, which are the polarity, arousal, emotion tag, emotion cues, and emotion source.

Q1: Polarity

Definition: An emotion polarity measures whether an emotion is pleasant or unpleasant. Is the person expressing positive, negative or no emotion? If the Polarity is None, it means that the sentence does not contain any emotion expression and you can skip the other emotion facets (arousal, emotion tag, emotion cues, and emotion source). If there are mixed polarities in a sentence, separate the polarity using a double semicolon (;;).

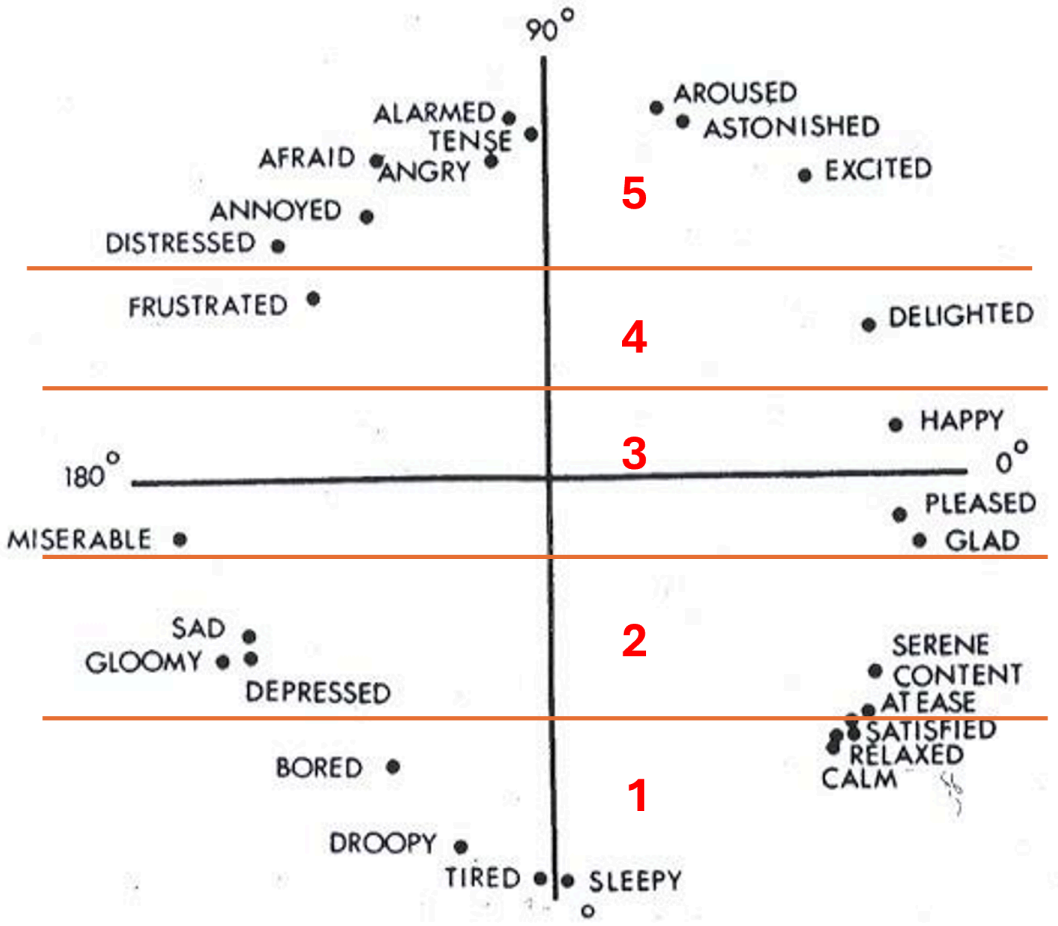
Code	Description
positive	Positive emotions are evoked by positive events, objects or situations. Emotions of pleasure (e.g. happiness, relaxed, fascination, love).
negative	Negative emotions are evoked by negative events, objects or situations. Emotions of displeasure (e.g., anger, fear, sadness, doubt).
neutral	Emotion expressed is neither positive nor negative (e.g. surprise).
none	No emotion expressed.

Q2: Arousal

Arousal measures the degree an emotion causes someone to be in a state of activation or arousal, which can range from being calm to excited.

Code	Description
1	Very low arousal (Hope)
2	Low arousal (Disappointment, Confidence, Exhaustion, Shame)
3	Moderate arousal (Admiration, Desperation, Gratitude, Curiosity, Pride)
4	High arousal (Fascination, Surprise)
5	Very high arousal (Fear, Hate, Love)

The chart below provides you with some guidelines on how to decide the level of arousal for an emotion. Make the best judgement based on overall emotion expressed in the sentence.



Q3: Emotion Tag

Emotion tag represents an emotion category with a set of distinctive features. There are 28 emotion tags for you to choose from. You can annotate a sentence with more than one emotion tag if the sentence contains multiple emotions. For sentences with multiple emotion, identify the primary emotion tag first, and then followed by the others. Separate the emotion tags with a double semicolon (;;). You are only allowed to suggest a new emotion tag if none of the 28 emotion tags apply.

Code	Description	Tweet Examples	Review Examples
admiration	Someone or something regarded as impressive or worthy of respect. Honoring or looking up to someone.	● I have respect for the boys who will stand up for their girls.	● i enthusiastically recommend this camera .
amusement	State of finding something funny or entertaining.	● Hilarious intro! well written: "How NOT To Go Up In Flames During A Social Media Crisis" via @AndyVale http://t.co/uy2gerrD9A	
anger	Feeling displeased, dissatisfied, annoyed, frustrated, hostile or rage caused by an undesirable event.	● Give me one reason why i shouldnt run up in mcdonalds wit a ak47 & get a killstreak for this disrespect @mcdonalds http://t.co/N5TDKO0ItV	● also , you need to have the headset plugged in all the time to have the radio work and that can get tedious .
boredom	Feeling dull, uninterested or left without anything in particular to do.	● @FilozofA ok bored of playing with little slavonic barbarian, #scumblock	
confidence	Feeling of self-assurance arising from one's appreciation of one's own abilities or qualities.	● I'm confident it won't be long now before human footprints follow in	● the day finally arrived when i

	Feeling one can trust or rely on someone or something.	the path of the Mars rover #Curiosity.	was sure i 'd leave sprint .
curiosity	Strong desire to know or learn something.	• I wonder if @cartoonnetwork would be interested in Fudge Lord http://t.co/tY5jA5UsAB	
desperation	Feeling complete loss of hope or despair, typically one that results in rash or extreme behavior. Suffering or driven by great need or distress.	• "#callmatterface I want to talk about how desperate Luca Modric is to leave Spurs he ransacked Tottenham last night."	
disappointment	Feeling displeasure caused by the non-fulfillment of one's hopes/expectations.		• when compared to the ipod the zen xtra falls short , but only by a bit .
disgust	Feeling that something is offensive, poisonous or contaminating that can be perceived with physical senses (sight, smell, touch, sound, taste), by the actions and appearances of people, and even by ideas.		• there are multiple photos in the ads of the tv “ in use “ but not a single one that shows that ugly black box !

Code	Description	Tweet Examples	Review Examples
doubt	State of being bewildered, confused, uncertain or unclear about something. Having mixed feelings about someone or something. Feeling of distrust, suspicion or one cannot rely on someone or something.	● i dont even know why im watching it tbh	
excitement	Feeling great enthusiasm and anticipation in considering some expected or longed-for good event.	● Go Giants! @SenFeinstein and I bet Sen. @Stabenow & @SenCarlLevin that we beat the Tigers. The stakes: http://t.co/dyw1beaj #WorldSeries	● the manual mode is feature rich and i can 't wait to get the hang of it .
exhaustion	State of physical or mental fatigue or feeling tired.	● I'm already exhausted from everything I'm going to avoid getting done today.	
fascination	State of being fascinated, amazed or interested in something. Feeling of great wonder or awe.	● Amazing middle students @SalkSchool in ElkRiver... learned about their #STEM program &saw projects firsthand. We should double stem schools!	● i 'd always eyed the nokia phones and had heard decent things about t-mobile , so i gave it a whirl .
fear	Feeling caused by the belief that someone or something is dangerous, likely to cause pain, or a threat. Feeling dread or anticipate with great apprehension or fear. Feeling anxious or worried over actual or potential problems.	● PAX panic is starting to hit me. I can't find my backpack. :|	
gratitude	State of being thankful or readiness to show appreciation for and to return kindness.	● Thank you to everyone who came	

		out to the Worcester rally to show their support. #peopleoverparty #masen http://t.co/oFZkdbOE	
happiness	Feeling pleased, satisfied, happy or delighted about a desirable event.	• Excellent news for Marylanders! MT @baltimoresun: BGE: Power has been restored to all customers affected by #MDSandy.	• im a more happier person after discovering the i/p button !
hate	Feeling of dislike, distaste or aversion towards a person, event or thing. Feeling of disgust or profound disapproval aroused by something unpleasant or offensive.	• and i really didnt like the character designs for cheria/sophie • @noahmittman @Gibbomadness I hate everything about it	• some things that i did n't like (but hey . nothing is perfect).

Code	Description	Tweet Examples	Review Examples
hope	Feeling of expectation and desire for a certain event to happen or grounds for believing something good will happen.	● Hoping the promised new direction for Doctor Who 'materialises' tonight, no more running, shouting, magic wand waving please.	● i hope it 's because it 's new and they are not out yet , but being without a remote is hard .
indifference	Lack of interest, concern, or sympathy.	● I know retweeting praise for yourself is frowned on, but what the hell... Just this once I don't care!	
inspiration	Feeling that makes someone want to do something or that gives someone an idea about what to do or create.	● Inspirational kindness from #CT's @NestleUSA donating half a million bottles of water for #Sandy victims (that's 12 truckloads!)	
jealousy	Feeling or showing envy of someone or their achievements and advantages. Feeling or showing suspicion of someone's unfaithfulness in a relationship.	● I would kill to have the body of the inzano twins ok	
longing	Yearning for or missing someone or something that one cannot have or cannot get easily. Feeling nostalgic, sentimental longing or wistful affection for the past, typically for a period or place with personal associations.	● Looking at a photograph and wishing you could re-live that moment over and over again. ● Miss that Louis https://t.co/rVyQ2Lbyi1	

love	Feeling of affection or natural liking towards another person, event or thing.	• I am in love 😍 http://t.co/vuSeaVUB2b • Doing my speech on the Olympics because I'm obsessed #Olympics #Sochi2014 #USAUSAUSA 🇺🇸🙌	• i fell in love with this camera , it combines ease of use , with an immense amount of options and power .
pride	Deep pleasure derived from one's own achievements, the achievements of those with whom one is closely associated, or from qualities or possessions that are widely admired.	• Proud to get a 100% score on hunger and nutrition issues from @FPAction: http://t.co/UZB4l3Jy	
regret	Feeling remorse or repentance over something that has happened or has been done. Feeling guilty of having done wrong or failed in an obligation.	• im so sorry i will never say that again omfg • @AlfieGallagher Sometimes I wish I did less to achieve more.	

Code	Description	Tweet Examples	Review Examples
relaxed	Feeling calm, at ease. Relief following release from anxiety or distress.	• A night to just lay up & watch movies on Netflix. • Lucky i didnt go to the spurs game last night	
sadness	Feeling of loss, helplessness or sorrow for own misfortune.	• It's sad that ppl try 2 hurt ppl they love because they refuse to be truthful. So they'd rather hurt you with "kill shots" and nasty fights • whAT THE IMCRYIBG SO HARD http://t.co/8bqRF6iLMk	
shame	Humiliation or embarrassment caused by the consciousness of wrong or foolish behavior. Feeling uncomfortable or awkward in a situation.	• "Yesterday is the day the atomic bomb in Japan. I embarrassed as a Japanese, was a moment of silence for the first time." • That awkward moment when your stalking someones instagram and you like a pic from 8238.3 weeks ago	
surprise	Unexpected or astonishing event, fact or thing. Sudden shocking event or experience.	• I CANT BELIEVE AT THE END BECAUSE LUDGER WAS IN FULL CHROMATUS WE COULDNT SEE HIS FACE	

		• @whatisrightt a week :O actually im surprised by how quick it went	
sympathy	Feeling of pity and sorrow for someone else's misfortune. Feeling empathy and expressing the ability to understand and share the feelings of another.	• My thoughts and prayers are with all those affected by hurricane #Sandy. • From Minnesota to Massachusetts: our hearts go out to the victims of today's tragedy. #prayforboston	

Q4: Emotion Cues

Identify the linguistic cues that influenced you to annotate the sentence with a particular emotion tag.

1) Copy and paste the portion of text you have identified as emotional cues into the Emotional Cues column in the annotation sheet.
2) There is no restriction to the length of each emotion cue. Emotion cues can be symbols, characters, words or phrases.
3) There can be multiple emotion cues associated with an emotion in a sentence. Identify all the cues you deem relevant to each emotion in a sentence. You can separate each cue for the same sentence with a comma (,).
4) If you have annotated a sentence with multiple emotion tags, make sure you specify which emotion tag the emotion cues are associated with.
    - Example: emo_tag_1: emo_cue_1.1, emo_cue_1.2;; emo_tag_2: emo_cue_2

Q5: Emotion Source

Identify whose emotion is being expressed or described.

Code	Description
self	Reviewer is expressing his or her own emotion.
other person	Reviewer is describing another person's emotion.
no one	Reviewer is describing an emotion-related phenomenon (e.g. general description of a particular emotion, trying to make someone feel certain emotion or talking about how someone should feel in a particular situation)

Section 2

The section below focuses on specific emotion tags, which are Ekman's six basic emotions (happiness, sadness, anger, surprise, fear, disgust) and two other emotions (disappointment, hate) that are relevant to the domain of product reviews.

Q6: Emotion Cause

Definition: An emotion cause refers to the specific event, situation or entity that directly triggers an emotional response in a person or character. It consists of an object (a specific entity or concept) and/or an action (an event, condition, or change involving the object). The object serves as the source or target of the action, while the action represents the dynamic aspect that affects the object. Together, the object and action form a complete cause that clearly explains why the emotion is felt.

* include do, is, has, can, should, could, would, will for emotion cause action

Sub-task: Identify the cause or reason that led to the tagged emotion in the sentence. Specify the object and the associated action. The text span should include the start and the end of the phrase, with each word separated by a space. For contractions, annotate based on actual spacing in the text.

1) Copy and paste the relevant portion(s) of the text into the Emotion Cause Object and Emotion Cause Action columns in the coding sheet.

2) There is no restriction to the length of each cause. The cause can be symbols, characters, words or phrases.

3) If no specific emotion cause is mentioned in the sentence (e.g. the sentence is very short and expresses only the emotion), annotate the cause object and cause action as "none".

4) If no specific action is mentioned in the sentence (e.g. the emotion cause only consists of an object), annotate the cause action as "none", and specify the cause object as the entity or concept responsible for the emotion.

5) Use specific separators to annotate multiple causes as follows:

- Use double semicolon (;;) to separate causes linked to different emotions.
- Use double ampersands (&&) to separate multiple cause objects and/or actions associated with the same emotion.
- Use double forward slashes (//) to separate one cause object with multiple associated actions under the same emotion.

Refer to the table below for examples of how to use these separators across different annotation scenarios.

Scenario	Emotion Cause Object	Emotion Cause Action
Emotion Label: 1 Cause Object: 1 Cause Action: 1	object_1	action_1
Emotion Label: 1 Cause Object: 1 Cause Action: 2	object_1	action_1.1.1 // action_1.1.2
Emotion Label: 1 Cause Object: 2 Cause Action: 1 for each cause object	object_1.1 && object_1.2	action_1.1 && action_1.2
Emotion Label: 1 Cause Object: 2 Cause Action: 2 for each cause object	object_1.1 && object_1.2	action_1.1.1 // action_1.1.2 && action_1.2.1 // action_1.2.2
Emotion Label: 2 Cause Object: 1 for each emotion label Cause Action: 1 for each cause object	object_1.1;; object_2.1	action_1.1;; action_2.1
Emotion Label: 2 Cause Object: 1 for each emotion label Cause Action: 2 for each cause object	object_1.1;; object_2.1	action_1.1.1 // action_1.1.2;; action_2.1.1 // action_2.1.2
Emotion Label: 2 Cause Object: 2 for each emotion label Cause Action: 1 for each cause object	object_1.1 && object_1.2;; object_2.1 && object_2.2	action_1.1.1 && action_1.2.1;; action_2.1.1 && action_2.2.1
Emotion Label: 2 Cause Object: 2 for each emotion label Cause Action: 2 for each cause object	object_1.1 && object_1.2;; object_2.1 && object_2.2	action_1.1.1 // action_1.1.2 && action_1.2.1 // action_1.2.2;; action_2.1.1 // action_2.1.2 && action_2.2.1 // action_2.2.2

Examples

Sentence	Emotion Tag	Emotion Cause Object	Emotion Cause Action
if you dont get video back , now you can run the player over with your car !	anger	player	dont get video back
im a more happier person after discovering the i/p button !	happiness	i/p button	discovering
i 've had the player for about 2 years now and it still performs nicely with the exception of an occasional wwhhhrrr sound from the motor .	happiness	player	still performs nicely
for the price it is a well spent investment !	happiness	price	investment
i 'm easily enlarging pictures to 8 1/2 x 11 with no visable loss in picture quality and not even using the best possible setting as yet (super fine) .	happiness; surprise	picture quality;; picture quality	no visable loss;; not even using the best possible setting as yet

Q7: Emotion Experiencer

Definition: An emotion experiencer is the person, character, or entity who feels the emotion expressed in a sentence. It is the subject or participant whose emotional state is being described or implied.

Sub-task: Identify who is experiencing the emotion described in the sentence. The text span should include the start and the end of the phrase, with each word separated by a space. For contractions, annotate based on actual spacing in the text.

1) Copy and paste the relevant portion of the text into the Emotion Experiencer column in the coding sheet.

2) There is no restriction on the length of the experiencer text.

3) If no specific experiencer is mentioned (e.g., the sentence discusses a general emotion phenomenon or how someone should feel), annotate the experiencer as “none”.

4) Use specific separators to annotate multiple experiencer as follows:

- Use double semicolon (;;) to separate experiencers linked to different emotions.
- Use double ampersands (&&) to separate multiple experiencers associated with the same emotion but different cause object.
- Use double forward slashes (//) to separate experiencers associated with the same emotion and same cause object but different cause action.

Examples

Sentence	Emotion Tag	Emotion Experiencer
if you dont get video back , now you can run the player over with your car !	anger	none
im a more happier person after discovering the i/p button !	happiness	im
i 've had the player for about 2 years now and it still performs nicely with the exception of an occasional wwhhhrrr sound from the motor .	happiness	i
for the price it is a well spent investment !	happiness	none
i 'm easily enlarging pictures to 8 1/2 x 11 with no visable loss in picture quality and not even using the best possible setting as yet (super fine) .	happiness;; surprise	i;; i
(he had owned two apex dvd players , and was pleased with both of them .)	happiness	he
after we discovered the problems , we tried to contact apex (fkor 3 months) , but we never got in touch with them.the customer support telephone number is always busy , and they do not answer their web customer support .	anger	we